

TASK 03 REPORT

INTERSHIP
CYBERSECURITY

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Week 3: Advanced Security and Final Reporting

1. Basic Penetration Testing

In the final week, I performed basic penetration testing on the application to simulate common attack scenarios. These tests helped evaluate the effectiveness of the security measures applied in Week 2.

- Attempted various attack patterns such as unauthorized access, broken authentication, and parameter tampering.
- Verified whether the implemented security mechanisms (like JWT middleware and input validation) were effectively preventing exploitation.

The application successfully resisted these basic penetration attempts, indicating improved resilience.

2. Logging and Monitoring

To enhance visibility into security-related activities, I added logging functionality using a logging library.

- **Login attempts** were recorded to help identify unusual or repeated access patterns.
- Error messages and failed login attempts were also logged for analysis.
- Logs were saved both on the console and in a dedicated security log file.

This setup can help in detecting intrusion attempts and performing audits if suspicious behavior is observed.

3. Creating a Security Best Practices Checklist

I compiled a simple checklist of best practices based on the security measures implemented during the internship:

- All user inputs are validated and sanitized
- Passwords are securely hashed and salted before storage
- Token-based authentication is used to secure APIs
- Helmet is applied to enforce security headers
- HTTPS is configured for encrypted communication
- Server does not expose unnecessary version information

- Developer comments and sensitive information are removed from production code
- Security logs are maintained for monitoring

Summary

Week 3 focused on **testing the effectiveness** of security controls and ensuring the system remained robust under potential attack attempts. Logging and monitoring capabilities were introduced, and a final checklist of best practices was created for long-term maintenance of application security.

The entire three-week process covered vulnerability identification, fixing issues, strengthening system defenses, and documenting progress — successfully fulfilling the goals of the internship project.